Geometry
Unit 13
Lesson 9 Practice

Name__Answer Key_______
Date___________________
Period_________________

Najira is using a computer software to design the layout of her bathroom remodel. The base of the bathroom creates $DEFG$ with coordinates $D(7,-2)$, $E(1,-6)$, $F(3,-9)$, and $G(9,-5)$.

1. Complete the table. Show your work in the table. Leave answers in exact form.

Line Segment	Length
$\overline{DE}$	$DE = \sqrt{(1-7)^2 + (-6+2)^2} = \sqrt{(-6)^2 + (-4)^2} = 2\sqrt{13}$
$\overline{EF}$	$EF = \sqrt{(3-1)^2 + (-9+6)^2} = \sqrt{13}$
$\overline{FG}$	$FG = \sqrt{(9-3)^2 + (-5+9)^2} = 2\sqrt{13}$
$\overline{DG}$	$DG = \sqrt{(9-7)^2 + (-5+2)^2} = \sqrt{13}$

2. Complete the table. Show your work in the table. Leave answers in exact form.

Diagonal Line Segment	Length
$\overline{DF}$	$DF = \sqrt{(3-7)^2 + (-9+2)^2} = \sqrt{65}$
$\overline{EG}$	$EG = \sqrt{(9-1)^2 + (-5+6)^2} = \sqrt{65}$

3. Finish the statement.

$DEFG$ is classified as a rectangle because … both pairs of opposite sides are congruent, diagonals are congruent.

4. If 1 unit on the coordinate plane equals 3 feet (ft), what are the dimensions of the base of the bathroom? Round to the nearest thousandth.
Length: $2\sqrt{13}(3) = 6\sqrt{13} \approx 21.633$ ft.
Width: $\sqrt{13}(3) = 3\sqrt{13} \approx 10.817$ ft.

5. $PRNT$ has vertices at $P(3,-8)$, $R(2,0)$, $N(-5,4)$, and $T(-4,-4)$.

$PRNT$ is classified as a(n) rhombus because... all sides are equal, diagonals are not equal.

$PR = RN = NT = TP = \sqrt{65}$

$PN = \sqrt{(-8)^2 + 12^2} = 4\sqrt{13}$

$RT = \sqrt{(-6)^2 + (-4)^2} = 2\sqrt{13}$

Ginny is designing a template for her new 3D printer. She created two quadrilaterals using her software.

$MSFC$	$TDPR$
$M(0,-7), S(4,-3), F(6,5), C(2,1)$	$T(5,7), D(7,4), P(0,4), R(2,7)$

6. Which quadrilateral can be classified as a trapezoid? Show your work.

$TDPR$

$TD = \sqrt{2^2 + (-3)^2} = \sqrt{13}$

$DP = \sqrt{(-7)^2 + 0^2} = 7$

$PR = \sqrt{2^2 + 3^2} = \sqrt{13}$

$RT = \sqrt{(-3)^2 + 0^2} = 3$

Slopes:

$\overline{TD}: m = -\frac{3}{2}$

$\overline{DP}: m = 0$

$\overline{PR}: m = \frac{3}{2}$

$\overline{RT}: m = 0$

7. List the properties of a trapezoid that you used.

One pair of opposite sides are parallel (same slope).

8. If 1 unit of the coordinate plane equals 4 inches (in.), what are the measurements of the sides of the trapezoid?

$TD = 4\sqrt{13} \approx 14.422$ in.

$DP = 28$ in.

$PR = 4\sqrt{13} \approx 14.422$ in.

$RT = 12$ in.

Fintan was graphing square $RDCP$, but mixed up his vertices. He knows that point R has coordinates $(4,-5)$ and point P has coordinates $(2,-2)$. His two options for points C and D are shown.

Point C	**Point D**
$(-1,-4)$	$(-1,-7)$
$(-1,-3)$	$(1,-7)$

9. Which coordinates are correct for points C and D? Justify your answer or show your work.

$RP = \sqrt{13}$, slope$=-\frac{3}{2}$

$RD = \sqrt{13}$, slope$=\frac{2}{3}$

$PC = \sqrt{13}$, slope$=\frac{2}{3}$

$C(-1,-4)$, $D(1,-7)$

10. Which properties of a square did you use to determine the correct coordinates for points C and D?

Length of all 4 sides are congruent and slope of adjacent sides opposite reciprocal (sides perpendicular).